ROBOCORE ZAMBIA

RoboCore Zambia

PLASTAL-BOT BUILDERS

GET STARTED HERE

www.plastalbotbuilders.com

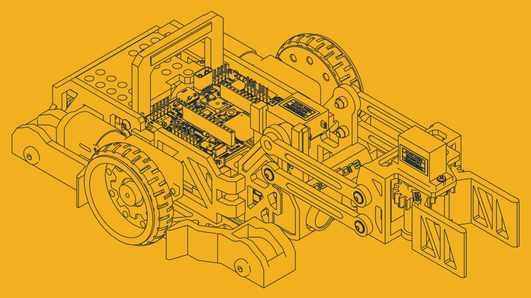
On this page you will find:

- **Assembly and first-run guide**
- **Step-by-step classroom lessons**
- **Programming environments: blocks, Python, C++**
- **Firmware, example code and libraries**
- **3D-print files for the frame, the lift and the gripper**
- **Support forum and teacher resources**

SCAN FOR DOCS, CODE AND LESSONS

BEFORE YOU START

- 3 x 18650 lithium cells (not included) and a USB-C charger
- A computer or tablet with a browser, or the Arduino IDE
- Charge the robot fully before the first lesson
- Raise the lift by hand before you switch the robot on



Replacement parts and add-on sensors:
plastalbotbuilders.com/robocore

Design files are published. Print it, fix it, improve it, share it.

PLASTAL-BOT BUILDERS

Plastal-Bot Builders is a Lusaka makerspace that teaches robotics, coding and fabrication to learners and teachers across Zambia. RoboCore is our own controller family, designed so a class can build, program and repair its robots locally.

WORKSHOPS · TEACHER TRAINING · ROBOTICS CLUBS · FABRICATION
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SCAN

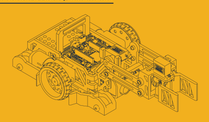
JOIN THE ROBOCORE COMMUNITY

Share what your class builds, find lesson plans and compare notes with other schools. Replacement parts and add-on sensors are available from us; the design files are published so you can also print and fix parts yourself.

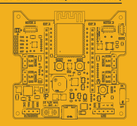
LESSON PLANS · FORUM · SPARE PARTS · DESIGN FILES
www.plastalbotbuilders.com/community
SCAN

THIS KIT INCLUDES

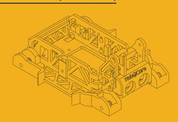
RoboCore robot, assembled



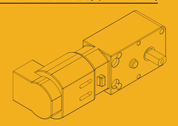
RoboCore V2 controller (on the robot)



3D-printed frame (on the robot)



Encoder motors DG01D-E (2, on the robot)



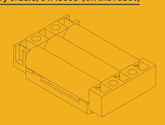
Wheels 56 mm, TPU tyres (2, on the robot)



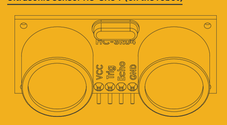
V-wheels 24 mm, with bearings (4, on the robot)



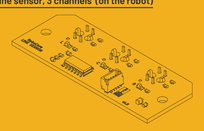
Battery cradle, 3 x 18650 (on the robot)



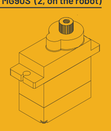
Ultrasonic sensor HC-SR04 (on the robot)



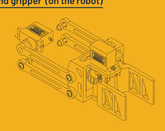
Line sensor, 3 channels (on the robot)



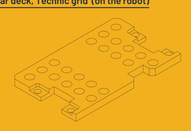
Micro servos MG90S (2, on the robot)



Lift and gripper (on the robot)



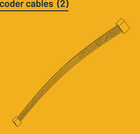
Rear deck, Technic grid (on the robot)



Carry handle (on the robot)



Motor encoder cables (2)



USB-C cable



Screws and quick-start card



For replacement parts and add-on sensors visit plastalbotbuilders.com/robocore

ROBOCORE · BUILD. CODE. CREATE.

Designed and built in Zambia by Plastal-Bot Builders, Lusaka

INSIDE (mirrored: prints on the back of the outside sheet). RoboCore mailer box, roll-end tuck-top, outer 303 x 213 x 91 mm, E-flute 1.5 mm. Print 1:1; navy solid = cut, grey dashed = fold. Bar = 100 mm.